

tube feet bear suckers. Calcareous ossicles of the body unite to form a rigid **test** or **corona** or **case**. Madreporite and anus are aboral in position. Ambulacral grooves are closed. Pedicellariae are three jawed. In the mouth of sea urchin a complex five jawed masticatory apparatus called **Aristotle's lantern** is present (absent in the heart urchins). Development includes **echinopluteus larva**.

Examples: *Echinus* (sea urchin), *Echinocardium* (heart urchin), *Echinodiscus* (sand dollar), *Clypeaster* (cake urchin).

Class-4: Holothuroidea (Gr. *holothurian* : water polyp or sea cucumber; *eidos*:form)

This class includes sea cucumbers. Body is elongated in the oro-aboral axis; body wall is leathery (coriaceous) and dermis contains loose spicules. Arms, spines and pedicellariae are absent. Mouth is surrounded by retractile tentacles (modified tube feet useful for feeding). Ambulacral grooves are 'closed'; tube feet bear suckers. Madreporite is *internal* (occurs in coelom). Respiratory organs are a pair of cloacal '**respiratory trees**'. Development is indirect and includes **auricularia larva**.

Examples: *Holothuria*, *Synapta*, *Thyone*.

3.10 PHYLUM - HEMICHORDATA

- Half chordates

3.10.1 General Characters

Hemichordata was earlier considered a sub-phylum under phylum Chordata. But now it is placed as a separate phylum under the non-chordates. This phylum consists of a small group of **worm-like** marine animals. They are **bilaterally symmetrical**, **triploblastic** and **coelomate (enterocoelous) animals**. The body is cylindrical and is composed of an anterior **proboscis**, a **collar** and a long **trunk** (Figure 3.31). A median **buccal diverticulum**, the **stomochord** (once believed to be the notochord) extends into the proboscis. Circulatory system is of open type with a dorsal heart. Respiration takes place through paired gill slits opening into the pharynx. Proboscis gland acts as the excretory organ. Development is indirect and includes a **tornaria larva**.

Examples: *Balanoglossus* (acorn worm), *Saccoglossus*.

3.10.2 Classification of the phylum Hemichordata

Phylum : Hemichordata is classified into **two classes**,

1. Enteropneusta and **2. Pterobranchia**.

Class-1: Enteropneusta (Gr. *enteron* : gut; *pneustos* : breathed)

They are commonly called *acorn worms*. They are free, solitary and burrowing animals. Development includes a free swimming **tornaria larva**.

Examples: *Balanoglossus* (acorn worm), *Ptychodera*.

Class-2: Pterobranchia (Gr. *pteron*: feather or wing; *branchion* : gill)

They are sedentary, colonial (*Rhabdopleura*), and tube dwelling animals.

Examples: *Rhabdopleura*, *Cephalodiscus*.

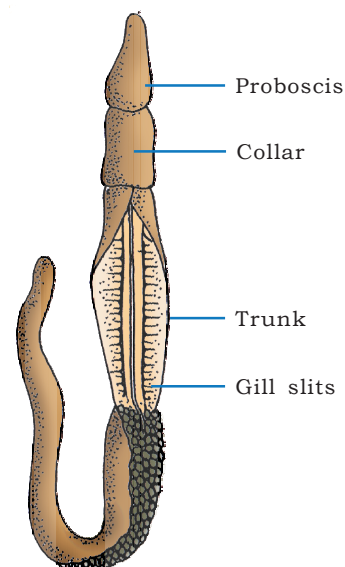


Fig. 3.31 Balanoglossus

GLOSSARY

Amphids: They are the cuticular depressions present on the lips surrounding the mouth in the nematodes. They are well developed in free living nematodes and serve as chemoreceptors.

Autotomy: It is the process of voluntary breaking of the injured part of the body (self mutilation or amputation). It is a device developed to protect the body from the enemies and parasites as seen in the echinoderms .

Bothridia: The leaf like organs present on the scolex of certain cestodes. They act as the organs of attachment to the body parts of the host.

Cercaria: It is a free swimming larva consisting of an ovoid body and a tail. It occurs in the life history of liver flukes.

Choanocyte: It is a specialized flagellated cell with a cup shaped collar around the base of the flagellum. It maintains the current of water through the body.

Clitellum: It is a belt or *girdle* shaped thickening of the skin, in a specific region of the body (14 to 16 segments in *Pheretima*). It secretes the cocoon and albumen (food for the developing young one), during the breeding season.

Cocoon: It is a bag like structure secreted by the clitellum. Eggs and sperms are deposited into it. Fertilization and development occurs within the cocoon.

Comb plates: They are ciliated plates helping in locomotion in the ctenophores.

Ctenidia: They are the respiratory organs of molluscs. Each ctenidium (gill) consists of a central axis and one or two rows of filaments.

Flame cell: It is a hollow cell containing a tuft of cilia in the inner space. It is excretory and osmoregulatory in function. Flame cell is a primitive type of excretory organ (protonephridium).

Gemmules: They are the '*internal buds*' of some sponges aiding in asexual reproduction. They also help tide over unfavourable conditions.

Juvenile: It is an immature young form resembling the adult in all aspects.

Madreporite: It is a circular porous plate present in the body of most echinoderms. Through its pores sea water enters the water vascular system.

Mesenteries: They are vertical partitions dividing the coelenteron of sea anemones. They are developed as inward folds of the endoderm. They bear cnidoblasts.

Miracidium: It is a free swimming larva of liver fluke. It contains ciliated epidermis, penetration glands, etc. It transforms into the sporocyst stage in the body of freshwater snail , which is the invertebrate host of liver fluke.

Moulting or Ecdysis: It is the process of casting off the outer body wall. During development of nematode larvae, the cuticle is cast off 4 times.

Muller's larva: It is the characteristic larva of turbellarians. It bears an apical tuft of cilia.

Nephridia: They are highly coiled excretory organs derived from the ectoderm.

Osculum: It is a large opening present at the free end of a sponge body. Water from the spongocoel flows out through the osculum.

Ostia: They are numerous openings present on the body of a sponge. Water is drawn in through the ostia and sent out through the osculum (choanocytes play an important role in drawing in water).

Parapodia: They are hollow biramous appendages of polychaetes. They bear tufts of many setae and help in locomotion.

Parenchymula: It is the larva of sponges such as *Clathrina*. It is similar to the planula of cnidarians. It is covered with cilia all over.

Pedicellariae: They are 'forceps like structures' (modified spines) in the body of sea stars and sea urchins. They remove foreign particles that may settle on the body surface and thus help in keeping the body surface clean.

Phasmids: They occur in some nematodes. They are well developed in parasitic nematodes and are glandulo-sensory in function.

Planula: It is a ciliated, double walled, free swimming larva characteristic of a cnidarian.

Polyembryony: Formation of several young ones from a single zygote. It occurs during the formation of redia and cercaria larvae in the life history of a liver fluke.

Polyp or Hydranth: It is a flower shaped or hydra shaped zooid of the colonial cnidarians such as *Obelia*. It bears several tentacles around the mouth. It is a nutritive zooid.

Radula: It is a thin ribbon shaped body lying in the buccal cavity of molluscs. It consists of a median stalk and two lateral lobes. Each lobe bears many rows of chitinous teeth. Radula works like a 'file' and rasps the vegetable food.

Redia: It is an elongated larva of liver fluke produced by the sporocyst asexually/ parthenogenetically. It gives rise to the second generation of rediae or the cercaria larvae.

Rhabdites: They are rod shaped bodies present in the epidermal cells of turbellarians. On discharge they form a slimy (mucus) substance helping in locomotion by smoothening the path of gliding movement and also in defence.

Setae: They are small, *f*-shaped, chitinous structures present in the pits (*setal sacs*) of the body wall of earthworms. They aid in locomotion. In some setae are modified into '*penial setae*' and help in copulation.

Spicules of a sponge: They are the supporting structures of the body secreted by *scleroblasts*. They are of different shapes and sizes. They are made up of calcium carbonate or silica.

Spongocoel: It is the large central cavity of a sponge body.

Statocyst: Statocyst is a sense organ concerned with the maintenance of equilibrium of the body.

Syncytium: It is a multinucleate mass of protoplasm without any division into cells. The tegument of flatworms is syncytial.

Trichemella: It is a larval form characteristic of the hexactinellid sponges.

Trochophore: It is the characteristic larva of annelids. It is 'top' shaped and bears one or more bands of cilia, an apical tuft, an eye etc. It contains coelom and nephridia. It is believed to be the forerunner of higher invertebrates.

Zooid: It is an individual member of a 'coelenterate' colony. Different types of zooids serve different functions (division of labour).

VERY SHORT ANSWER TYPE QUESTIONS

1. Name the cells and system peculiar to porifera.
2. Which structures form the skeleton of a sponge? What are the chemicals involved in the formation of these structures?
3. What are the functions of canal system of sponges?
4. What are the two chief morphological 'bodyforms' of cnidarians? What are their chief functions?
5. What is metagenesis? Animals belonging to which phylum exhibit metagenesis?
6. What is the cnidarian group with quantitatively/relatively large mesoglea? What is the significance of such a well developed mesoglea pertaining to the aquatic life of that group?
7. What is the chief difference between the hydrozoans and the rest of the cnidarians regarding the germinal layer(s) in which its 'defensive structures or cells of defence' occur?
8. What are the excretory cells of flatworms called? What is the other important function of these specialized cells?
9. Distinguish between amphids and phasmids.
10. What is the essential difference between a 'flat worm' and a 'round worm' with reference to the perivisceral area of their bodies?
11. How do you account for the origin of the perivisceral space in the body of a nematode and an annelid?
12. What is metamerism? What is the essential difference between the mode of formation of individual morphological body units of a tapeworm and those of an earthworm?
13. How do you distinguish a 'hirudinean' from the rest of the annelids, based on the morphological features pertaining to metamerism? How does the coelom of a leech differ from the coelom of an earthworm with reference to its contents?
14. What do you call the locomotor structures of *Nereis*? Why is *Nereis* called a polychaete?
15. What is botryoidal tissue?
16. What is the difference between the epidermis of a nematode and that of an annelid? How does a nematode differ from an annelid with reference to the musculature of the body wall?
17. What do you call the first and second pairs of cephalic appendages of a scorpion?
18. What is the uniqueness about the first two pairs of cephalic appendages of a crustacean compared to those of the other extant arthropods?
19. What is the sub-phylum to which 'ticks' and 'mites' belong? How do you distinguish them from the insects with reference to their walking legs?
20. What are the respiratory structures of *Limulus* and *Palamnaeus* respectively?
21. What are antennae? What is the arthropod group without antennae?
22. What do you call the perivisceral cavity of an arthropod? Where from is it derived during development?
23. Which arthropod, you have studied, is called a 'living fossil'? Name its respiratory organs?
24. How do you identify a *Chiton* from its external appearance? How many pairs of gills help in the respiration of *Chiton*?

25. What is the function of radula? Give the name of the group of molluscs which do not possess a radula?
26. What is the other name for the gill of a mollusc? What is the function of osphradium?
27. What is Aristotle's lantern? Give one example of an animal possessing it?
28. What is the essential difference between the larvae and adults of echinoderms, symmetry wise?
29. What are blood glands in *Pheretima*?
30. What are spermathecae on the body of *pheretima*?

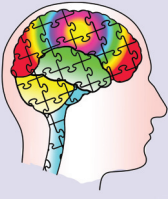
SHORT ANSWER TYPE QUESTIONS

1. Write short notes on the salient features of the anthozoans.
2. What is the class to which the flukes belong? Write short notes on the chief characters of that group?
3. What are the salient features exhibited by polychaetes?
4. How do the hirudineans differ from the polychaetes and oligochaetes?
5. What are the chief characters of the crustaceans?
6. Mention the general characters of Arachnida.
7. Compare briefly a centipede and a millipede.
8. Cephalopods show several unique or advanced features when compared to the other molluscs. Discuss briefly.
9. What are the salient features of pelecypoda.
10. What are the salient features of the echinoids?

11. Mention the salient features of Holothuroidea.
12. What is the function of nephridia?
13. Give an account of the hearts in the circulatory system of *Pheretima*.
14. Name the phylum to which comb jellies belong to and mention its important features.
15. Give a brief account of hemichordates.

LONG ANSWER TYPE QUESTIONS

1. Draw a labelled diagram of the reproductive organs of *Pheretima*.
2. Describe the digestive system and process of digestion in *Pheretima*.



For ignited minds

- Not For Evaluation

1. Why do we call *Limulus*, a 'Living fossil'?
2. How do you distinguish a '**King crab**' from a '**scorpion**' based on a morphologically distinctive character, on the ventral aspect of the prosomas of the two animals.
3. When there is no respiratory pigment in the blood of an insect or a centipede or a millipede, why should there be a pigment in the blood of a scorpion?
4. Blood flows in our body. Similarly blood flows in the body of cockroach also (in spite of the open circulatory system, it has). How come that the blood of cockroach is mostly referred to as 'haemolymph'?
5. There is a perivisceral cavity in the body of a nematode almost similar to that seen in the body of an earthworm. In such a case, why do you call it a 'pseudocoel' instead of 'coelom'.
6. "All non chordates are 'schizocoelomates' and chordates are 'enterocoelomates'. Do you agree with the above statement? Give reasons for your positive or negative answer.
7. "All schizocoelomates are protostomeates, while all protostomeates are not schizocoelomates". Do you think the statement is correct? If so give reasons. If not, how do you defend.
8. Echinoderms show 'pentamerous radial symmetry'. Yet, they are classified under the Grade Bilateria. Does it not look anomalous to you? What is your suggestion to taxonomists? Do you want them to reclassify the animal world?
9. In almost all animals of the animal kingdom, gonads arise from the germinal layer called mesoderm. However, some animals /animal groups differ. Can you name them and explain the origin of germ cells in them.
10. There is a maximum number of four chambers in the heart of a vertebrate. There are invertebrates with more number of atria and ventricles. Can you give examples for such organisms.